

I A S 2 0 2 6 · O R A L P R E S E N T A T I O N

Foot Core vs Trunk Core Training in CAI

Postural Control Strategies · 10-minute talk script

Full script, June's voice — Brian models it first, then it becomes your practice text.

/ 짧은 쉬임 · // 긴 쉬임 · 골드 = 강세 · 구간별 목표 시간 표기

Good Morning

~0:45

Good **morning**, everyone. //

My name is Jiyeon **Lee**, / from **Yonsei** University. //

Today I'd like to talk about / two ways of training the **core** — //

from the **foot**, / and from the **trunk** — //

and what they do / to movement **strategies** //

in people with chronic **ankle** instability. //

Here's the question / I want to leave with you: //

🔑 if the **ankle** is unstable, //

where should we **start** — / from the **bottom**, / or from the **top**? //

Why This Matters

~1:45

Let me start with the **problem**. //

After an ankle **sprain**, / many people develop //

chronic ankle instability — / or **C-A-I**. //

Their balance is **poorer**, //

and their whole **body** moves differently / to make up for it. //

We call these / **compensatory** strategies. //

Now — core stability **matters** here. //

And in C-A-I, / **two** cores are weak: //

the **foot** core, / down at the bottom — //

and the **trunk** core, / up at the top. //

During balance tasks, / these patients show //

altered movement / in the **legs** / and in the **trunk**. //

The body is / **working** around / the unstable ankle. //

But here's what we **don't** know: //

🔑 if we train from the **bottom** up, / or from the **top** down — //

can we actually **change** / those strategies? //

That was our **question**. //

What We Did

~2:15

So here's what we **did**. //

We recruited **thirty** individuals with C-A-I, //
following the International Ankle Cons**ort**ium criteria, //
and randomly split them / into **three** groups: //
foot core training, / **trunk** core training, //
and a **control** group / with no intervention. //

The training groups completed / **twelve** supervised sessions — //
thirty **min**utes each, / three times a week, / for **four** weeks. //
We tested everyone / at **baseline** / and after the inter**vention**. //

Two **tasks**: //
single-leg **stance**, / and single-leg **jump** landing. //

For **static** control, / we measured C-O-P velocity / and sway **area**. //
For **dynamic** control, / we measured D-P-S-I //
and the kin**em**atics / of the legs and trunk — //
at initial **contact**, / and in the first two hundred mill**is**econds after. //

🔑 For the continuous data, / we used statistical parametric **mapping** — //
so we could see the **whole** landing phase, / not just single points. //

What We Found

~2:15

Now — what did we **find**? //

First, the honest part: //

for **most** postural control outcomes — / static and dynamic — //

there were **no** significant interactions. //

Four weeks of training / did **not** change / the overall balance numbers. //

But — / and this is the interesting part — //

during the single-leg **jump** landing, //

we found significant interactions / in **trunk** rotation: //

at initial contact, / p equals point-oh-one-**six** — //

and at peak angle, / p equals point-oh-oh-**four**. //

And here's the surprise. //

🔑 It was **not** the trunk training group. //

Only the **foot** core group / reduced their trunk rotation — //

compared with **both** other groups. //

The S-P-M analysis confirmed it: //

across the **entire** two-hundred-millisecond window, //

the foot core group showed / the biggest **improvement** //

in reducing excessive trunk **rotation**. //

What It Means

~1:45

So what does this **mean**? //

Remember the opening question — //

bottom, / or **top**? //

Our data says: / start from the **bottom**. //

When the **foot** becomes stable, //

the trunk no longer **needs** / to compensate so much. //

That's a bottom-up / **kinetic** chain effect: //

fix the found**ation**, //

and the movement above it / **calms** down. //

For rehabilit**ation**, / the message is practical: //

🔑 adding **foot** core training / to C-A-I protocols //

may be a highly effective way / to change **dynamic** movement patterns — //

even if the static **sway** / stays the same. //

Static balance is one thing. //

But **landing** — / the moment injuries actually **happen** — //

that's where foot core training / made the **difference**. //

Thank You

~0:45

To **sum** up: //
four weeks of **foot** core training //
reduced compensatory **trunk** rotation / during landing — //
something trunk training itself / did **not** do. //

Sometimes the answer to a problem **up** here / ...
 ...starts **down** there. //

Thank you very much. //
I'd be happy to take / any **questions**. //